This assignment will not be graded and is only for practice.

Level 1

1) Which of these sets form an orthonormal basis of $\mathbb{R}^3$ with respect to the dot product as the inner product in $\mathbb{R}^3$? ***1 point***

- [ ] $\{(1,0,0),(0,1,0)\}$
- [ ] $\{(2,0,0),(0,2,0),(0,0,2)\}$
- [ ] $\{(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}),(\frac{-3}{\sqrt{11}},\frac{1}{\sqrt{11}},\frac{1}{\sqrt{11}}),(\frac{1}{\sqrt{66}},\frac{7}{\sqrt{66}},\frac{-4}{\sqrt{66}})\}$
- [ ] $\{(2,1,-1),(-1,-1,1),(1,2,-2)\}$
- [ ] $\{(\frac{1}{\sqrt{2}},0,\frac{-1}{\sqrt{2}}),(0,1,0),(\frac{1}{\sqrt{2}},0,\frac{1}{\sqrt{2}})\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}),(\frac{-3}{\sqrt{11}},\frac{1}{\sqrt{11}},\frac{1}{\sqrt{11}}),(\frac{1}{\sqrt{66}},\frac{7}{\sqrt{66}},\frac{-4}{\sqrt{66}})\}$
$\{(\frac{1}{\sqrt{2}},0,\frac{-1}{\sqrt{2}}),(0,1,0),(\frac{1}{\sqrt{2}},0,\frac{1}{\sqrt{2}})\}$

2) Consider two orthogonal vectors a, b in $\mathbb{R}^2$. If $a+b$ and $a-b$ are orthogonal, then choose the correct option. ***1 point***

- ( ) $\|a\| = \|b\| = 1$
- ( ) $\|a\| = \|b\|$
- ( ) $\|a\| = 2\|b\|$
- ( ) $2\|a\| = \|b\|$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\|a\| = \|b\|$

3) ***1 point***

Consider $a = (1, 1), b = (1, -1)$. Let $V = Span\{a, b\}$. Choose the correct options by considering the standard inner product (dot product).

- [ ] Vectors a, b form an orthogonal basis for V.
- [ ] Vectors a, b form an orthonormal basis for V.
- [ ] There exist scalar multiples of a, b which form an orthonormal basis
- [ ] There do not exist scalar multiples of a, b which form an orthonormal basis

No, the answer is incorrect.

Score: 0

Accepted Answers:

Vectors a, b form an orthogonal basis for V.
There exist scalar multiples of a, b which form an orthonormal basis

Level 2

4) Choose the set of correct statements. ***1 point***

- [ ] The determinant of a matrix formed by 3 orthonormal vectors in R^3 is ± 1.
- [ ] The determinant of a matrix formed by 3 orthonormal vectors in R^3 is 0.
- [ ] The determinant of a matrix formed by 2 orthonormal vectors in R^2 is ± 1.
- [ ] The determinant of a matrix formed by 2 orthonormal vectors in R^2 is 0.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The determinant of a matrix formed by 3 orthonormal vectors in R^3 is ± 1.
The determinant of a matrix formed by 2 orthonormal vectors in R^2 is ± 1.

5) Consider a system of linear equations: ***1 point***

$$2x_1 + 2x_2 + 7x_3 = b_1$$

$$2x_1 + x_2 - 10x_3 = b_2$$

$$3x_1 - 2x_2 + 2x_3 = b_3.$$

Let A be the coefficient matrix of the given system of linear equations. Let a matrix B contain the column vectors of A, which are normalized by their respective norms, as its columns (i.e. first column vector of A normalized by its norm is the first column of B). Which of the following statements are true ?

- [ ] The determinant of BB^T is 1.
- [ ] BB^T is an identity matrix.
- [ ] BB^T is a scalar matrix.
- [ ] BB^T is a diagonal matrix.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The determinant of BB^T is 1.
BB^T is an identity matrix.
BB^T is a scalar matrix.
BB^T is a diagonal matrix.

6) Choose the correct option(s). ***1 point***

- [ ] The vectors in an orthonormal set are linearly independent.
- [ ] A set of linearly dependent vectors can be orthonormal.
- [ ] A set of linearly independent vectors is always orthonormal.
- [ ] A set of linearly independent vectors is always orthogonal but not orthonormal.
- [ ] The vectors in an orthogonal set are linearly independent.

No, the answer is incorrect.
Score: 0

Accepted Answers:

The vectors in an orthonormal set are linearly independent.
The vectors in an orthogonal set are linearly independent.

Consider the inner product $\langle u, v\rangle = 3u_1v_1 + 2u_2v_2$ where $u = (u_1, u_2), v = (v_1, v_2)$ are vectors in $\mathbb{R}^2$. Answer questions 7 and 8 based on this information.

7) Which of the following is an orthonormal basis with respect to the inner product defined above? ***1 point***

- $\{(2,3), (-4,4)\}$
- $\{\frac{1}{\sqrt{30}}(2,3), \frac{1}{\sqrt{80}}(-4,4)\}$
- $\{\frac{1}{\sqrt{13}}(2,3), \frac{1}{\sqrt{32}}(-4,4)\}$
- $\{(2,3), (-3,2)\}$
- $\{\frac{1}{\sqrt{13}}(2,3), \frac{1}{\sqrt{13}}(-3,2)\}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\{\frac{1}{\sqrt{30}}(2,3), \frac{1}{\sqrt{80}}(-4,4)\}$

8) Use the orthonormal basis $\{u, v\}$ obtained in question 7 with respect to the defined inner product. Express the vector $(4, 0)$ as a linear combination of the basis vectors u and v, as $(4,0) = c_1u + c_2v$. Which of the following gives the coefficients of the linear combination? ***1 point***

- $c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}}$
- $c_1 = \frac{24}{\sqrt{13}}, c_2 = \frac{-48}{\sqrt{32}}$
- $c_1 = \frac{8}{\sqrt{13}}, c_2 = \frac{-16}{\sqrt{32}}$
- $c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{48}{\sqrt{80}}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}}$

Your score is: 0/8